

POWER_FEATURES_PORTING_PLAN

POWER FEATURES PORTING PLAN — CLI-Agent Power Features → HelixCode

Field	Value
Revision	2
Created	2026-06-14
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Status	DRAFT — research + plan only, NO code (operator decision pending). Rev 2: Phase-1 status reconciled against HEAD e7266263 — T1.1/T1.2/T1.3 found already DONE; T1.4/T1.5/T1.6 PARTIAL; stale paths corrected (internal/autonomy→internal/workflow/autonomy, internal/planner→internal/workflow/planmode).
Authority	§11.4.70 subagent-driven, §11.4.74 catalogue-first, §11.4.6 no-guessing, §11.4.50/§11.4.98 anti-bluff
Method	Read-only research of cli_agents/<agent>/ READMEs+docs; cross-checked against helix_code/internal/* + helix_code/applications/*
Scope	Map distinctive power features of every in-repo CLI agent for porting into HelixCode

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1. Executive Summary

- **Agents enumerated under `cli_agents/`:** 51 directories. Of these, ~38 are **coding/agent tools** worth mining; the remainder are MCP servers (`git-mcp`, `postgres-mcp`, `conduit`, `noi`, `snow-cli`, `x-cmd`, `cli-agent`, `aiagent`, `superset`, `ui-ux-pro-max`, `fauxpilot`, `deepseek-cli*`, `xela-cli`, `codex-skills`) or a skills/plugin marketplace (`bridle` = “Tons of Skills”) rather than full agents.
 - **Distinctive power features mapped:** ~140 across the 24 agents researched in depth (`aider`, `cline`, `plandex`, `crush`, `codex`, `gemini-cli`, `qwen-code`, `copilot-cli`, `amazon-q`, `mistral-code`, `codename-goose`, `gptme`, `open-interpreter`, `forge`, `gpt-engineer`, `swe-agent`, `shai`, `nanocoder`, `claude-squad`, `claude-code`, `vtcode`, `warp`, `get-shit-done`, `spec-kit`, `junie`, `aichat`, `codai`, `taskweaver`, `octogen`, `zeroshot`, `multiagent-coding`, `agent-deck`). After de-duplicating equivalent features across agents, they collapse into ~62 **distinct capabilities**.
 - **HelixCode is already mature.** It HAS: agentic tool loop (`internal/agent`), read-only tools + browser suite (`internal/tools`), MCP client+lifecycle (`internal/mcp`), plugins (`internal/plugins`), skills (via plugins + `crush/vtcode-style SKILL.md` precedent), LSP-equipped context, ensemble + verifier (`internal/verifier`), `repomap` (`internal/repomap`, `tree-sitter`), sessions + history-condense (`internal/session`), hooks (`internal/hooks`), sandbox + worktrees (`internal/worker`, `internal/session/manager_worktree`), memory + cognee (`internal/memory`), voice-to-code (`internal/voice`, the `aider P2-F27` port), clarification (`internal/clarification`), task checkpoints (`internal/task/checkpoint.go`), roocode delegation (`internal/roocode`), kilocode callgraph/impact (`internal/kilocode`), planner/plantree (`internal/planner`, `internal/plantree`), approval/approvalwire, `autocommit`, `git_status`, multi-platform apps (`desktop`/`terminal-ui`/`ios`/`android`/`aurora`/`harmony`).
 - **Net result:** Most “power features” are **HAVE** or **PARTIAL** — the real porting work is in **autonomy presets, cumulative-diff sandbox review, conversation branching, messaging connectors, daemon/server-shared-session mode, spec-driven workflow surface, and tangent/rewind UX**. Few features are genuinely **MISSING/large**.
 - **Phasing:** 8 phases (0–7), ordered high-value-low-effort first, ending in niche features. Every feature ships with the operator’s full anti-bluff bar (§7).
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2. Method & Evidence Discipline

Per §11.4.6 (no-guessing) and §11.4.74 (catalogue-first), every feature row cites the source-agent file where the capability is textually evidenced. HelixCode status was cross-checked by reading `helix_code/internal/<pkg>/doc.go`, function signatures, and the existing port plan `docs/plans/HXC-031-codex-cline-port.md`. No source was modified. Where the in-repo doc only links out to a vendor website (`codex docs/* .md` stubs, `mistral-code`’s inherited `gemini` docs), that is noted so we do not over-claim a feature exists in-tree.

Status legend: HAVE (shipped in HelixCode) · PARTIAL (substrate exists, surface/UX gap) · MISSING. **Effort legend:** S (small, ≤300-line unit) · M (medium, 1–3 units) · L (large, multi-component).

3. What HelixCode Already Has (baseline)

Verified by reading the inner module (`helix_code/internal`, `helix_code/applications`):

Capability	HelixCode location	Evidence
Multi-agent orchestration + agent types	<code>internal/agent</code>	<code>agent/doc.go</code> (Coordinator, WorkflowExecutor, planning/coding/debug/review agents)
Repo-map (tree-sitter, ranked, incremental)	<code>internal/repomap</code>	<code>repomap/file_ranker.go</code> , <code>cache.go</code> , <code>incremental_p2t06_test.go</code>
MCP client + lifecycle + chaos-tested	<code>internal/mcp</code>	<code>mcp/lifecycle.go</code> , <code>config.go</code> , <code>mcp_chaos_test.go</code>
Plugins (loader, activation, registry, exec)	<code>internal/plugins</code>	<code>plugins/loader.go</code> , <code>activation.go</code> , <code>registry.go</code>
Ensemble + verifier (LLMsVerifier single source)	<code>internal/verifier</code> , <code>internal/llm</code>	<code>verifier/client.go</code> , <code>adapter.go</code> ; ensemble commits in git L
Sessions + history compaction/condense	<code>internal/session</code>	<code>session/condense.go</code> (CompactIfNeeded, ShouldCompa <code>manager_worktree.go</code>
Hooks (lifecycle blockers/executor)	<code>internal/hooks</code>	<code>hooks/hook.go</code> , <code>blockers.go</code> , <code>executor.go</code>
Sandbox + worktree isolation	<code>internal/worker</code> , <code>internal/session</code>	<code>worker/isolation_test.go</code> , <code>session/manager_worktr</code>
Memory + cognee + project memory	<code>internal/memory</code> , <code>internal/projectmemory</code>	<code>memory/manager.go</code> , <code>cognee_integration.go</code> , <code>project loader.go</code>
Voice-to-code (aider port P2-F27)	<code>internal/voice</code>	<code>voice/doc.go</code> (Whisper API + whisper.cpp fallback)
Clarification (LLM-generated questions)	<code>internal/clarification</code>	<code>clarification/engine.go</code> (DetectAmbiguity, Resolve

Capability	HelixCode location	Evidence
Task checkpoints (DB-backed)	internal/task	task/checkpoint.go (CreateCheckpoint/GetLatestCheckpoint/restore)
Auto-commit + secret-filter + summariser	internal/autocommit	autocommit/committer.go, secret_filter.go, summariser.go
Roo-code delegation, Kilo-code callgraph/impact	internal/roocode, internal/kilocode	roocode/delegator.go, kilocode/callgraph.go, impact.go
Planner / plan-tree (compact, operations)	internal/planner, internal/plantree	planner/executor.go, plantree/operations.go
Approval engine + yes/no prompter	internal/approval, internal/approvalwire	approval/manager.go, approvalwire/yesno_prompter.go
Browser tools (chromedp, click/type/screenshot)	internal/tools	tools/browser*_v2.go
Quality gate + scorer	internal/quality	quality/gate.go, scorer.go
Multi-platform apps + TUI	applications/*	applications/{desktop,terminal_ui,ios,android,aurora_os,harmattan}
Multimodal (in-flight per HXC-031)	internal/llm/vision, HXC-031 plan	docs/plans/HXC-031-codex-cline-port.md

This baseline means the plan is **mostly closing UX/surface gaps over existing substrate**, not building from zero.

4. Agents Surveyed

Full coding agents mined (24): aider, cline, plandex, crush, codex, gemini-cli, qwen-code, copilot-cli, amazon-q, mistral-code, codename-goose, gptme, open-interpreter, forge, gpt-engineer, swe-agent, shai, nanocoder, claude-code, vtcoder, aichat, codai, taskweaver, octogen.

Orchestration / multi-agent managers (6): claude-squad, agent-deck, zeroshot, multiagent-coding, warp, get-shit-done.

Workflow / spec toolkits (2): spec-kit, junie.

Not full agents (noted, low-priority mining): `bridle` (skills/plugin marketplace), `git-mcp/postgres-mcp/conduit/noi/snow-cli/x-cmd/cli-agent/aiagent/superset/ui-ux-pro-max/fauxpilot/deepseek-cli*/xela-cli/codex-skills` (MCP servers / dashboards / stubs).

Confirmed ABSENT from `cli_agents/`: `openhands` (operator example) is NOT present — do not plan against it.

5. Master Feature Matrix

feature × representative source-agent(s) × HelixCode status × effort × priority (1=highest). Evidence column cites the canonical source file.

#	Feature	Source agent(s)	Evidence (<code>cli_agents/...</code>)	HelixCode status
F01	Repo-map (tree-sitter project map)	aider, plandex, codai	<code>aider/aider/website/docs/repomap.md</code> ; <code>plandex/README.md</code> ; <code>codai/README.md</code>	HAVE (integrated)
F02	Git auto-commit w/ generated message	aider, plandex, forge	<code>aider/README.md</code> ; <code>plandex/docs/.../version-control.md</code> ; <code>forge/README.md</code> (:commit)	HAVE (integrated)
F03	Architect/editor dual-model mode	aider	<code>aider/aider/website/docs/usage/modes.md</code>	PARTIAL (explicit architecture)
F04	Chat modes (code/ask/architect/plan)	aider, plandex, gemini-cli, forge	<code>aider/.../modes.md</code> ; <code>plandex/.../repl.md</code> ; <code>gemini-cli/docs/cli/plan-mode.md</code> ; <code>forge (sage/muse)</code>	PARTIAL (planned first-class mode)
F05	Voice-to-code	aider	<code>aider/.../voice.md</code>	HAVE (integrated)
F06	Lint & test on edit (auto-fix loop)	aider, cline, swe-agent	<code>aider/.../lint-test.md</code> ; <code>cline/README.md</code> ; <code>swe-agent/docs/background/aci.md</code>	PARTIAL (quality as per-edit lint)
F07	<code>/undo + /diff</code> (git-aware)	aider, forge	<code>aider/.../commands.md</code>	HAVE (DONE: <code>main.go:2</code> internal/ <code>git.go:85</code>)
F08	Images & web pages in chat (<code>/web</code> , <code>/paste</code>)	aider, gpt-engineer	<code>aider/.../images-urls.md</code> ; <code>gpt-engineer/README.md</code> (--image_directory)	PARTIAL (Helix flight; <code>/web</code>)
F09	Watch-files / IDE-comment mode	aider	<code>aider/.../watch.md</code>	MISSING
F10	Copy/paste-to-web-chat (<code>/copy-context</code>)	aider	<code>aider/.../copypaste.md</code>	MISSING

#	Feature	Source agent(s)	Evidence (cli_agents/...)	HelixCode status
F11	Plan/Act dual mode	cline, gemini-cli	cline/README.md; gemini-cli/docs/cli/plan-mode.md	HAVE (DONE internal/mode_control/wired cmd/c
F12	Checkpoints (workspace snapshot + restore/undo)	cline, gemini-cli, amazon-q, nanocoder	cline/README.md; gemini-cli/docs/checkpointing.md; amazon-q/docs/experiments.md; nanocoder/docs/features/index.md	PARTIAL (ta workspace-fi
F13	MCP marketplace + on-the-fly tool creation	cline, codename-goose	cline/README.md; goose extension system	PARTIAL (M marketplace/a
F14	.clinerules/GEMINI.md/ AGENTS.md context files	cline, gemini-cli, codex, forge, shai, mistral-code	cline/.clinerules/*; gemini-cli/GEMINI.md; codex/AGENTS.md; shai/README.md (SHAI.md)	HAVE (inte rules)
F15	Plugin SDK (programmatic tools + hooks)	cline, gptme	cline/README.md (@cline/sdk); gptme/README.md	HAVE (inte
F16	Multi-agent teams (coordinator → specialists)	cline, codename-goose, multiagent-coding, zeroshot	cline/README.md; goose GooseTeam; multiagent-coding/README.md	PARTIAL (ag delegation; ne
F17	Scheduled/cron agents	cline, nanocoder	cline/README.md; nanocoder daemon (croner)	MISSING
F18	Messaging connectors (Slack/Telegram/Discord/...)	cline	cline/README.md	MISSING (H exists org-wi
F19	Headless / non-interactive (JSON/stream-json out)	cline, gemini-cli, qwen-code, codex, shai, vtcode, gpt-engineer	cline/README.md; gemini-cli/docs/cli/headless.md; codex/docs/exec.md; shai/README.md; vtcode/README.md	PARTIAL (c JSON/stream
F20	Background long-process monitoring	cline, plandex	cline/README.md; plandex/.../execution-and-debugging.md	PARTIAL (i background
F21	Kanban / parallel task board	cline, agent-deck	cline/README.md; agent-deck/llms.txt	MISSING

#	Feature	Source agent(s)	Evidence (cli_agents/...)	HelixCode status
F22	Cumulative diff-review sandbox (stage-before-apply)	plandex, claude-squad, forge	plandex/.../reviewing-changes.md; claude-squad/README.md; forge/README.md (--sandbox)	PARTIAL (w cumulative-d
F23	Configurable autonomy levels (preset tiers)	plandex, nanocoder, copilot-cli	plandex/.../autonomy.md; nanocoder modes; copilot-cli Autopilot	HAVE (DON internal/modes.go:
F24	Automated command/build/test debugging	plandex, shai, gptme	plandex/.../execution-and-debugging.md; shai/README.md (shell hook); gptme	PARTIAL (ve debug loop on
F25	Conversation branches / fork	plandex, forge, claude-squad, agent-deck	plandex/.../branches.md; forge/README.md; agent-deck/llms.txt	MISSING (se fork)
F26	Big-context management (load files/dirs/urls; smart context)	plandex, gemini-cli, nanocoder	plandex/.../context-management.md; gemini-cli/README.md; nanocoder File Explorer	HAVE/PART repomap; @-n partial)
F27	Model packs / mid-session model switch	plandex, crush, copilot-cli, junie, qwen-code	plandex model packs; crush/README.md; copilot-cli/model; qwen-code/model	HAVE (LLM aliases.g
F28	LSP-enhanced context	crush, copilot-cli, nanocoder	crush/README.md; copilot-cli/README.md; nanocoder (source/lsp/)	HAVE (LSP r
F29	Permission allowlist + --yolo/ auto-approve	crush, vtcode, open-interpreter, claude-squad, qwen-code	crush/README.md; vtcode/README.md; open-interpreter/README.md (-y)	HAVE (inte
F30	Provider auto-update catalogue	crush (Catwalk)	crush/README.md	HAVE (LLM CONST-036)
F31	Shared workspace across clients (server backend)	crush, qwen-code (daemon), shai (serve)	crush/README.md; qwen-code/README.md; shai/README.md	MISSING (se session multi
F32	Self-config skill (“configure yourself”)	crush	crush/README.md (crush-config skill)	PARTIAL (co driven skill)
F33		codex, gemini-cli,	codex/AGENTS.md; codex/docs/sandbox.md;	

#	Feature	Source agent(s)	Evidence (cli_agents/...)	HelixCode status
	OS sandbox for tool exec (Seatbelt/Landlock/bwrap)	octogen, taskweaver	octogen/README.md; taskweaver container	PARTIAL (w/ OS-level seatbelt confirmed)
F34	Exec-policy / command-approval engine	codex, amazon-q, vtcode	codex/docs/execpolicy.md; amazon-q/docs/built-in-tools.md	HAVE (integrated selector)
F35	ACP mode (Agent Client Protocol over stdio)	gemini-cli, gptme, qwen-code	gemini-cli/docs/cli/acp-mode.md; gptme/README.md; qwen-code daemon (ACP)	MISSING
F36	/rewind (conversation + file-state rewind)	gemini-cli	gemini-cli/docs/cli/rewind.md	PARTIAL (cli only)
F37	Model routing / auto-fallback on quota	gemini-cli, crush	gemini-cli/docs/cli/model-routing.md	HAVE (integrated auto_llm_mode)
F38	Model steering (mid-task course-correct)	gemini-cli	gemini-cli/docs/cli/model-steering.md	MISSING
F39	Git worktrees per session	gemini-cli, claude-squad, forge, zeroshot, agent-deck	gemini-cli/docs/cli/git-worktrees.md; claude-squad/README.md	HAVE (integrated manager_worktrees)
F40	Agent Skills (SKILL.md open standard)	gemini-cli, crush, vtcode, forge, nanocoder, codex, qwen-code	gemini-cli/docs/cli/skills.md; crush/README.md; vtcode/README.md; forge/README.md	PARTIAL (2024-01-25 subsystem shared commands/2-tier project : 253-254; SKILL.md f
F41	ask_user / TODO / tracker tools	gemini-cli, amazon-q, nanocoder	gemini-cli/docs/tools/{ask-user,todos,tracker}.md; amazon-q/docs/experiments.md	PARTIAL (2024-01-25 DONE integrated askuser/a wired cmd/c main.go:7 background-j
F42	Knowledge base (persistent semantic index)	amazon-q, aichat (RAG), gptme (RAG), forge (semantic search)	amazon-q/docs/knowledge-management.md; aichat/README.md; forge/README.md (:sync)	HAVE (integrated RAG)
F43		amazon-q		MISSING

#	Feature	Source agent(s)	Evidence (cli_agents/...)	HelixCode status
	Tangent mode (side-topic checkpoint)		amazon-q/docs/tangent-mode.md	
F44	Introspect tool (self-aware about own features)	amazon-q	amazon-q/docs/introspect-tool.md	MISSING
F45	Delegate (async background sub-sessions)	amazon-q, gptme (subagent), nanocoder, vtcode	amazon-q/docs/experiments.md; gptme/README.md; vtcode/README.md	PARTIAL (no workflow back)
F46	Recipes (reusable parametrized workflows)	codename-goose, forge (skills)	goose recipes; forge/README.md	PARTIAL (pl parametrized)
F47	Lessons system (auto-injected contextual guidance)	gptme	gptme/README.md	MISSING
F48	Computer use (GUI/desktop control)	gptme, open-interpreter, warp (Oz)	gptme/README.md; open-interpreter/README.md	PARTIAL (br control)
F49	Local code-exec REPL (NL → run code)	open-interpreter, gptme, octogen, taskweaver	open-interpreter/README.md; octogen/README.md	HAVE (comm BLUFF-003 r)
F50	Profiles (YAML behavior presets)	open-interpreter, codename-goose, claude-squad	open-interpreter/README.md; goose profiles	PARTIAL (co behavior-prof)
F51	Whole-project generation from a prompt	gpt-engineer	gpt-engineer/README.md	PARTIAL (ag <dir> one-s
F52	Benchmark harness (SWE-bench/ TerminalBench)	gpt-engineer, swe-agent, codename-goose, multiagent-coding	gpt-engineer/README.md; swe-agent/README.md; goose goose-bench	MISSING (h benchmark)
F53	Agent-Computer Interface tools (windowed editor/viewer, succinct search)	swe-agent	swe-agent/docs/background/aci.md	PARTIAL (ec viewer UX ab
F54	Shell assistant (failed-command auto-fix hook)	shai, aichat, plandex	shai/README.md; aichat/README.md	MISSING

#	Feature	Source agent(s)	Evidence (cli_agents/...)	HelixCode status
F55	Chainable --trace (pipe conversation between runs)	shai	shai/README.md	MISSING
F56	HTTP server (OpenAI-compatible endpoints)	shai, aichat, gptme, open-interpreter, octogen	shai/README.md; aichat/README.md; gptme/README.md	PARTIAL (i OpenAI-com)
F57	Context compression /compact + /usage	nanocoder, qwen-code, forge	nanocoder/docs/features/index.md; qwen-code/README.md	HAVE (inte condense.)
F58	Session auto-save + /resume	nanocoder, codex, vtcode	nanocoder/docs/features/index.md; vtcode/README.md	HAVE (inte
F59	Event triggers (file-watch + cron → skill)	nanocoder	nanocoder/CLAUDE.md (daemon)	MISSING
F60	VS Code / IDE extension bridge	nanocoder, codex, qwen-code, junie	nanocoder/docs/features/vscode-extension.md; codex/README.md	MISSING (or app team)
F61	Spec-driven workflow (specify→plan→tasks→implement)	spec-kit, get-shit-done	spec-kit/README.md; get-shit-done/docs/FEATURES.md	PARTIAL (pl command sur
F62	QA / regression gates in workflow	get-shit-done	get-shit-done/docs/FEATURES.md	HAVE (quali helix_qa)
F63	Cross-AI peer review (multi-runtime review)	get-shit-done, multiagent-coding	get-shit-done/docs/FEATURES.md	PARTIAL (en runtime review
F64	Token/usage tracking per request + context-%	codai, amazon-q, nanocoder	codai/README.md; amazon-q/docs/experiments.md	HAVE/PART prompt % inc
F65	Stateful code-first execution (DataFrames persist)	taskweaver, octogen	taskweaver/README.md; octogen/README.md	PARTIAL (su stateful)
F66	MCP socket pool (share MCP procs across sessions)	agent-deck	agent-deck/llms.txt	MISSING
F67	Global conversation search (fuzzy + regex)	agent-deck	agent-deck/llms.txt	MISSING
F68	Conversation export (: dump JSON/HTML)	forge, aichat	forge/README.md	MISSING
F69	Restricted/secure shell mode (path/symlink guards)	forge, vtcode	forge/README.md; vtcode/README.md (layered security)	PARTIAL (ap

#	Feature	Source agent(s)	Evidence (cli_agents/...)	HelixCode st
F70	ChatGPT/account-plan auth (non-API-key sign-in)	codex, junie, copilot-cli	codex/docs/ authentication.md; copilot-cli/README.md	OUT-OF-SCO verifier owns

6. Top-10 Highest-Value Missing Features

Ranked by (user impact × low duplication-with-existing × evidence strength). “Missing” or “Partial-surface” only — excludes things HelixCode already HAS.

1. **F22 — Cumulative diff-review sandbox (stage-before-apply, apply/reject hotkeys)** — plandex/forge/claude-squad. The single biggest trust/UX lever; HelixCode has worktrees but no cumulative-diff staging+review surface. **L.**
2. **~~F23 — Configurable autonomy levels (named preset tiers None→Full) — DONE~~** **2026-06-14** (internal/workflow/autonomy/modes.go:12-31, 5 named presets). No longer a missing feature; kept here only for ranking history.
3. **F12 — Workspace checkpoints (file snapshot + restore/undo)** — cline/gemini/amazon-q/nanocoder. HelixCode checkpoints are task-DB rows, not workspace-file snapshots; this is the “oops, revert everything” safety net. **M.** (Phase 2 — not re-verified this 2026-06-14 pass.)
4. **F11/F04 — Plan/Act + first-class chat modes (code/ask/architect/plan)** — cline/aider/gemini/forge. **F11 Plan/Act is DONE 2026-06-14** (internal/workflow/planmode/mode_controller.go:13-14, wired cmd/cli/main.go:773-779); F04’s broader code/ask/architect mode menu remains the open surface. **M.**
5. **F61 — Spec-driven workflow surface (specify→clarify→plan→tasks→implement + constitution)** — spec-kit/get-shit-done. Aligns directly with HelixCode’s constitution discipline. **L.**
6. **F25 — Conversation branches / fork** — plandex/forge/agent-deck. Explore multiple approaches/models from a shared point. **M.**
7. **F36/F43 — /rewind + Tangent mode** — gemini-cli/amazon-q. Side-explore and time-travel without losing place. **M.**
8. **F19/F56 — Robust headless (JSON/stream-json) + OpenAI-compatible server** — cline/gemini/shai/aichat. Unlocks CI/CD + programmatic embedding. **M.**
9. **F33 — OS-level tool-exec sandbox (Seatbelt/Landlock/bwrap)** — codex/gemini. Hard isolation beyond worktrees; safety-critical for full-auto. **L.**
10. **F18 — Messaging connectors (Slack/Telegram/Discord/Linear)** — cline. Org already has the Herald precedent to extend (§11.4.74 catalogue-first), making this lower-risk than greenfield. **L.**

Runners-up (just outside top-10): F35 ACP mode, F45 async delegate, F59 event triggers, F54 shell-assistant auto-fix.

7. Per-Feature Acceptance Bar (anti-bluff)

Per the operator's bar and §11.4.27 / §11.4.69 / §11.4.83 / §11.4.98 / §11.4.107 / §11.4.135, **every** ported feature is DONE only when it ships ALL of:

1. **Real implementation** — no simulation/placeholder (§Rule 2, BLUFF-001/002/003); touches real files/processes/LLM calls.
 2. **A triggering prompt** — the exact natural-language prompt that exercises the feature end-to-end (§11.4.105 intent-recognition compatible).
 3. **A recorded video/transcript demo** under docs/qa/<run-id>/ (§11.4.83) — full bidirectional thread + captured evidence.
 4. **Tests** — unit + integration + (where applicable) stress + chaos (§11.4.85), each anti-bluff with a paired §1.1 mutation; integration/E2E hit real infra (§11.4.50(A)).
 5. **A Challenge** in challenges/ exercising the complete user workflow (§Rule 6/7).
 6. **A HelixQA bank entry** executed in an autonomous session with captured wire evidence (§11.4.27 (B)).
 7. **A standing regression guard** registered on close (§11.4.135) with RED-on-broken-artifact polarity switch (§11.4.115).
 8. **Docs in sync** — feature manual/guide + tracker row with Catalogue-Check: field (§11.4.74).
-

8. Phases → Tasks → Sub-Tasks

Each task names concrete file targets (under helix_code/internal/... or helix_code/cmd/cli/...) and acceptance criteria. Tasks are sized for §11.4.70 subagent-driven, ≤~300-line units. Every sub-task inherits the §7 acceptance bar.

Phase 0 — Foundations & Audit

Goal: replace assumptions with FACTS before any port. No new features.

- **T0.1 — Capability-confirmation audit.** Verify the PARTIAL rows by reading code, not guessing (§11.4.6).
 - ST0.1.1 — Confirm headless JSON/stream-json output state in cmd/cli/main.go + internal/render. Acceptance: documented yes/no with file:line.
 - ST0.1.2 — Confirm SKILL.md precedence vs plugins in internal/plugins (F40). Acceptance: precedence table captured.
 - ST0.1.3 — Confirm OS-sandbox depth in internal/worker/isolation + internal/infraboot (F33). Acceptance: “worktree-only” vs “seatbelt/landlock” determination.
 - ST0.1.4 — Confirm OpenAI-compat surface in internal/server (F56). Acceptance: endpoint inventory.
- **T0.2 — Catalogue-check sweep (§11.4.74).** For each planned feature, record reuse | extend | no-match <org/repo@sha> (e.g. F18 → extend Herald; F42 → reuse cognee). Acceptance: every Phase 1–6 feature has a Catalogue-Check: line.
- **T0.3 — Tracker rows.** Open one Feature item per planned feature in the workable-items DB (§11.4.93) with created_by/assigned_to (§11.4.104). Acceptance: DB[?]MD in sync (workable-items validate).

Phase 1 — High-Value / Low-Effort (already-partial)

Wrap existing substrate into discoverable surfaces. Mostly S/M.

⚠ **Reconciled 2026-06-14 against HEAD e7266263** (read-only grep/read of the real inner tree, file:line cited per §11.4.6). Phase-1 was STALE: most items the plan filed as “to do / partial” are already SHIPPED — several under different package paths than the plan named, and T1.1’s CLI wiring (plan said “pending”) is in fact live. Per-item ground-truth status table is below; corrected paths are inlined. Future-phase content (Phases 2–7) was NOT re-verified this pass.

Item	Status (2026-06-14)	Evidence (file:line)	What was wrong in the original plan
T1.1 / undo+/ diff	DONE	handlers helix_code/cmd/cli/ main.go:2118-2133 (dispatch) + :2326-2371 (handleDiff/ handleUndo); real git via helix_code/internal/ autocommit/git.go:85 DiffUnstaged, :136 RevertLastCommit, :151 DiffSinceRef	Plan said substrate-only “no / undo//diff UX” and prior audit said “CLI wiring still pending” — both WRONG; both commands are fully wired and use real git output.
T1.2 Autonomy presets	DONE	helix_code/internal/ workflow/autonomy/ modes.go:12-31 (5 named modes None/Basic/BasicPlus/SemiAuto/ FullAuto), :137 Level() 1-5, capability table :51-122	Plan named the package internal/autonomy; it actually lives at internal/ workflow/autonomy/.
T1.3 Plan/ Act modes	DONE	helix_code/internal/ workflow/planmode/ mode_controller.go:13-14 (ModePlan/ModeAct), transition table :118-121; wired helix_code/cmd/cli/ main.go:773-779 (NewModeController, NewToolGate, enter/exit plan-mode tools)	Plan named internal/ planner for the mode controller; the first-class Plan/Act controller is internal/workflow/ planmode/ and is already CLI- wired.
T1.4 TODO/ ask_user surface	PARTIAL	ask_user DONE: helix_code/ internal/tools/askuser/ ask_user_tool.go:47-61, wired helix_code/cmd/cli/ main.go:707-714. /tasks exists (helix_code/internal/ commands/ tasks_command.go:29, wired main.go:769) but wraps workflow.BackgroundManager (background-job list), NOT the plantree/clarification TODO-tracker this item describes.	Plan implied the TODO surface bridges internal/ plantree+internal/ clarification; the shipped / tasks is a background-job tracker. The plantree slash command is plantree (internal/commands/ plan_tree_command.go:20), not a TODO /tasks.

Item	Status (2026-06-14)	Evidence (file:line)	What was wrong in the original plan
T1.5 Token + context-%	PARTIAL	per-request token count DONE: helix_code/cmd/cli/main.go:1727-1740 (printGenerationStats, real resp.Usage.PromptTokens/CompletionTokens, anti-bluff “don’t fabricate”). Context-% indicator ABSENT — no production hit for any context-window-percent render in internal/render or cmd/cli.	Plan filed F64 as “HAVE/PARTIAL”; the token half is DONE, the context-% half is genuinely ABSENT (only the token counter ships).
T1.6 Skill SKILL.md precedence	PARTIAL	markdown-skills subsystem ships: helix_code/internal/commands/markdown_skills.go (SkillLoader :217, precedence :253-254 “user first, then project; project overrides user”), wired helix_code/cmd/cli/main.go:991-1010 (.helix/skills/ + user-config dir).	NOT absent. Gaps vs the AC: (a) filename is any *.md (markdown_skills.go:267), not the SKILL.md open-standard filename; (b) precedence is 2-tier (project>user) — the built-in/bundled tier is MISSING (no //go:embed, no default skills). So “repo/user/built-in precedence” is only 2 of 3 tiers.

- **T1.1 — /undo + /diff (F07, S).  DONE (2026-06-14).** Shipped: handlers at cmd/cli/main.go:2118-2133 + :2326-2371, bridged to real git in internal/autocommit/git.go (DiffUnstaged:85, RevertLastCommit:136, DiffSinceRef:151). /undo does git revert --no-edit HEAD (force-push-free per §11.4.113); /diff prints verbatim git diff [<ref>]. Remaining (if AC tightening desired): the dedicated reproduction Challenge reverting a real edit. (NOTE: the prior audit’s “CLI wiring still pending” is corrected to DONE.)
- **T1.2 — Autonomy presets (F23, M).  DONE (2026-06-14).** Shipped at internal/workflow/autonomy/ (NOT internal/autonomy — path corrected). modes.go:12-31 defines the five named presets (None/Basic/BasicPlus/SemiAuto/FullAuto), Level():137 maps them to levels 1-5, and GetCapabilities:51-122 drives per-tier behaviour. AC residual: confirm the paired §1.1 mutation (flip a tier → test FAILs) is registered.
- **T1.3 — First-class chat/Plan-Act modes (F04, F11, M).  DONE (2026-06-14).** Shipped at internal/workflow/planmode/ (NOT internal/planner — path corrected). mode_controller.go:13-14 (ModePlan/ModeAct), legal transitions :118-121; CLI-wired at cmd/cli/main.go:773-779 via NewModeController + NewToolGate + enter/exit-plan-mode tools (tool availability gated by mode). AC residual: the “no writes in ask/plan mode” Challenge.
- **T1.4 — TODO/tracker tool surface (F41, S).  PARTIAL (2026-06-14).** ask_user DONE (internal/tools/askuser/ask_user_tool.go:47-61, wired cmd/cli/main.go:707-714). The plan’s TODO-tracker (plantree/clarification → /tasks) is NOT what shipped: /tasks (internal/commands/tasks_command.go:29, wired main.go:769) lists **background jobs** via workflow.BackgroundManager, not a step-by-step TODO tracker. Remaining: wire internal/plantree (slash command currently named plantree, internal/commands/plan_tree_command.go:20) + internal/clarification into a user-visible task-tracker surface.

- **T1.5 — Token/usage + context-% indicator (F64, S). ① PARTIAL (2026-06-14).** Per-request token count DONE: `cmd/cli/main.go:1727-1740` (`printGenerationStats`) prints `real resp.Usage.PromptTokens/CompletionTokens` from the provider usage object (anti-bluff: refuses to fabricate when unpopulated, §11.4.141-compliant). Remaining: the **context-window-% indicator is ABSENT** — no production code renders a context-usage percentage; this is the genuine outstanding work for T1.5.
- **T1.6 — Skill SKILL.md precedence (F40, S, gated on ST0.1.2). ① PARTIAL (2026-06-14).** A markdown-skills subsystem already ships (`internal/commands/markdown_skills.go + skills_command.go + skills_watcher.go`, wired `cmd/cli/main.go:991-1010`, dirs `.helix/skills/ + user-config-dir`). It implements **2-tier** precedence (project overrides user, `markdown_skills.go:253-254`). Remaining vs the AC: (a) the **built-in/bundled tier** is missing (no `//go:embed` defaults) → “repo/user/built-in” is only 2 of 3 tiers; (b) the loader keys on any `*.md` filename (: 267), not the SKILL.md open-standard filename. Document/implement the third tier + filename convention.

Phase 2 — Session & Context Power Tools

- **T2.1 — Workspace checkpoints (F12, M).** Targets: `internal/session + worktree` to snapshot/restore working-tree files (not just task DB). AC: `create→edit→restore` round-trips real files; chaos test mid-write SIGKILL recovers cleanly.
- **T2.2 — Conversation branches/fork (F25, M).** Targets: `internal/session` branch primitive + `cmd/cli/branch`. AC: fork from a point, diverge, compare; both branches persist.
- **T2.3 — /rewind (F36, M).** Targets: compose T2.1 checkpoints + history rewind in `internal/session/condense.go`. AC: rewind reverts conversation + optionally files.
- **T2.4 — Tangent mode (F43, M).** Targets: `internal/session` tangent stack. AC: enter tangent, explore, return to exact prior state; `tangent tail` keeps last Q+A.
- **T2.5 — Big-context @-mention + file explorer (F26, M).** Targets: `internal/context + repomap`; `@file/@dir/@url` loaders + token estimates. AC: `@-load` injects real content with token budget shown.
- **T2.6 — Conversation export + global search (F68, F67, S each).** Targets: `internal/session`: dump JSON/HTML; fuzzy+regex search across sessions. AC: exported HTML opens; search finds a known phrase.

Phase 3 — Autonomy & Safety Surface

- **T3.1 — OS-level exec sandbox (F33, L, gated on ST0.1.3).** Targets: `internal/worker/internal/infraboot` — Seatbelt (macOS), Landlock/bwrap (Linux), per §11.4.81 cross-platform parity (per-OS branches + honest kernel-gap citation). AC: a write outside the sandbox is blocked on each OS branch; paired mutation strips a branch → gate FAILs.
- **T3.2 — Lint/test-on-edit auto-fix loop (F06, M).** Targets: `internal/editor + internal/quality` — block edit on syntax/lint failure, auto-retry. AC: a deliberately broken edit is blocked then fixed; SWE-agent-style linter-block evidenced.
- **T3.3 — Automated command/build/test debugging (F24, M).** Targets: `internal/verifier` + new debug loop on failing exec. AC: a failing build is auto-diagnosed + fixed with captured evidence.
- **T3.4 — Shell-assistant failed-command hook (F54, M).** Targets: `cmd/cli/shell hook + internal/approval`. AC: a failed shell command yields an LLM-proposed fix (opt-in).

- **T3.5 — Restricted/secure shell mode (F69, M).** Targets: extend `internal/security` path/symlink/dangerous-command guards. AC: a path-escape command is blocked; audit log captured.

Phase 4 — Multi-Agent & Parallel Orchestration

- **T4.1 — Cumulative diff-review sandbox (F22, L).** Targets: new `internal/changeset` staging layer over worktrees + `cmd/cli` review menu (apply/reject per-file/hunk). AC: changes accumulate unapplied; `diff --ui` apply/reject lands only accepted hunks; reject leaves tree clean.
- **T4.2 — Persistent multi-agent teams (F16, L).** Targets: extend `internal/agent` coordinator + `internal/roocode` with named-team persistence. AC: coordinator delegates to specialists with isolated context; team state survives restart (§11.4.119 single-resource-owner for shared files).
- **T4.3 — Async delegate / background sub-sessions (F45, M).** Targets: `internal/workflow/background.go` + `internal/roocode`. AC: background sub-session runs in parallel, summary injected into main context (§11.4.89 background execution).
- **T4.4 — Cross-AI peer review (F63, M).** Targets: `internal/verifier` ensemble → multi-runtime reviewer. AC: a second model/runtime reviews a diff; findings captured.
- **T4.5 — Kanban/parallel task board + MCP socket pool (F21, F66, L).** Targets: app/server task board; `internal/mcp` shared-process pool. AC: parallel cards each on own worktree; MCP procs shared with measured memory reduction.

Phase 5 — Connectors, Server & Headless Surface

- **T5.1 — Robust headless output (F19, M, gated on ST0.1.1).** Targets: `cmd/cli --output-format json|stream-json`. AC: NDJSON event stream; `git diff | helix --json` works in a pipe; `-count=3` deterministic (§11.4.98).
- **T5.2 — OpenAI-compatible server (F56, M, gated on ST0.1.4).** Targets: `internal/server` chat-completions/responses/embeddings + SSE. AC: a real OpenAI client hits the endpoint and gets a completion.
- **T5.3 — Shared-session daemon (F31, L).** Targets: `internal/server` multi-client shared session keyed by cwd (crush/qwen/shai pattern). AC: two clients mirror the same session live; loopback no-auth, remote token-gated.
- **T5.4 — ACP mode (F35, M).** Targets: `cmd/cli --acp` JSON-RPC over stdio. AC: a JSON-RPC initialize/prompt round-trips (Zed/JetBrains-compatible shape).
- **T5.5 — Messaging connectors (F18, L, extend Herald per §11.4.74).** Targets: connector adapters → session mapping with access control. AC: a real Slack/Telegram thread drives a session with @-attribution (§11.4.104) + intent recognition (§11.4.105).
- **T5.6 — Chainable --trace (F55, S).** AC: `helix ... | helix "now run it"` pipes a real conversation trace.

Phase 6 — Spec-Driven & Quality-Gate Workflow

- **T6.1 — Spec-driven command surface (F61, L).** Targets: `cmd/cli` + `internal/planner/plantree` — `/specify,/clarify,/plan,/tasks,/implement,/analyze` with a project constitution file. AC: a spec produces a plan → tasks → real implementation; constitution governs phases. (Strong synergy with HelixCode's existing constitution discipline.)
- **T6.2 — Recipes (parametrized reusable workflows) (F46, M).** Targets: `internal/plugins/skills` — parametrized recipe primitive. AC: a recipe runs with substituted parameters end-to-end.

- **T6.3 — Event triggers (file-watch + cron → skill/recipe) (F59, F17, M).** Targets: new `internal/triggers` daemon (file-watch + cron) firing skills. AC: a file change fires a subscribed skill; a cron schedule survives restart.
- **T6.4 — Whole-project generation one-shot (F51, M).** Targets: `cmd/cli` scaffold mode reading a prompt file → full project. AC: `helix gen <dir>` from a prompt file produces a buildable project.

Phase 7 — Long-Tail / Niche Features

- **T7.1 — Watch-files / IDE-comment mode (F09, M).** AC: an `AI !` comment in a file triggers an edit.
- **T7.2 — Lessons system (auto-injected guidance) (F47, M).** AC: a keyword/tool match injects a lesson; opt-out works.
- **T7.3 — Introspect tool + self-config skill (F44, F32, S each).** AC: “what can you do?” answered from bundled docs; “configure yourself” edits config.
- **T7.4 — Model steering mid-task (F38, M).** AC: a steer message redirects without restarting the loop.
- **T7.5 — Profiles (YAML behavior presets) (F50, S).** AC: `--profile` switches a named behavior set.
- **T7.6 — Copy-context to web chat (F10, S).** AC: `/copy-context` yields paste-ready context.
- **T7.7 — Benchmark harness (F52, L) + ACI windowed editor (F53, M).** AC: agent runs against a public benchmark task; windowed viewer shows ~100 lines/turn with in-file search.
- **T7.8 — Computer use / desktop control (F48, L).** AC: a GUI action is driven + verified (compose §11.4.107 liveness).
- **T7.9 — Stateful code-first execution (DataFrames persist) (F65, L) + token-economy review** — only if a data-analytics use case is greenlit.

9. Dependency Touch-Map

Which org submodules / internal subsystems each feature cluster touches (per §11.4.51 equal-codebase, extend-don't-reimplement):

Feature cluster	HelixAgent	LLMsVerifier	HelixMemory (cognee)	HelixLLM	HelixSpecifier	Helix...
Modes/Autonomy (F04,F11,F23)	✓ agent loop	✓ model status	—	✓ routing	—	—
Checkpoints/Rewind/Tangent/Branches (F12,F25,F36,F43)	✓ session	—	partial (history)	—	—	—
Diff-sandbox (F22)	✓ worktree	—	—	—	—	—
Multi-agent/Delegate (F16,F45,F63)	✓ coordinator/roocode	✓ ensemble	✓ shared mem	✓	—	—
Server/Headless/ACP/Connectors (F19,F31,F35,F56,F18)	✓	✓	—	✓	—	✓ ex
		—	—	—		—

Feature cluster	HelixAgent	LLMsVerifier	HelixMemory (cognee)	HelixLLM	HelixSpecifier	H
Spec-driven (F61) + Recipes/Triggers (F46,F59)	✓ planner/ plantree				✓ (spec engine)	
Sandbox/Lint-fix/ Auto-debug (F33,F06,F24)	✓ worker/ editor	✓ verifier	—	—	—	—
Context/@-mention/ RAG (F26,F42)	✓ context	—	✓ cognee	—	—	—

HelixSpecifier is the natural owner for F61’s spec engine (catalogue-check before greenfield). Herald is the catalogue-first base for F18 messaging connectors.

10. Honest Effort Caveats

- **No over-claiming “missing”.** Many marquee features (repo-map F01, auto-commit F02, voice F05, model routing F37, worktrees F39, RAG/knowledge F42, condense F57, sessions F58, approval F29/F34, LSP F28, provider catalogue F30) are **already HAVE** in HelixCode — porting work there is zero or surface-only.
- **PARTIAL ≠ trivial.** F12/F22/F33/F61 have substrate but the user-facing surface + cross-platform + anti-bluff coverage are the real cost (each is M–L).
- **Vendor-doc stubs.** codex docs / *.md and mistral-code’s inherited gemini docs link out to websites; their feature claims were taken from README/AGENTS/crate-names, not in-tree prose — do not assume the in-repo copy is the spec.
- **Out-of-scope rows.** F60 (IDE extension), F70 (account-plan auth), F52 (agent benchmark) are flagged but are app-team / provider / research concerns, not core-CLI ports.
- **openhands absent.** The operator’s example list named openhands; it is NOT in cli_agents / — excluded.
- This is a **planning document** (§11.4.73 spec-versioning applies once accepted). No code written. Effort estimates are S/M/L bands, not hour commitments.

11. Sources Verified

Researched 2026-06-14 by read-only subagents against the in-repo vendored copies under cli_agents/. Primary evidence files (non-exhaustive):

- **aider:** aider/README.md, aider/aider/website/docs/{repomap,usage/modes,usage/commands,usage/voice,usage/watch,usage/lint-test,usage/copypaste,usage/images-urls}.md
- **cline:** cline/README.md, cline/.clinerules/*
- **plandex:** plandex/README.md, plandex/docs/docs/core-concepts/{plans,reviewing-changes,context-management,autonomy,execution-and-debugging,version-control,branches}.md, plandex/docs/docs/repl.md
- **crush:** crush/README.md, crush/docs/hooks/
- **codex:** codex/README.md, codex/AGENTS.md, codex/docs/{sandbox,exec,skills,slash_commands,config,agents_md,execpolicy,authen
codex/codex-rs/ crate layout

- gemini-cli: gemini-cli/README.md, gemini-cli/docs/cli/{plan-mode,rewind,model-routing,model-steering,git-worktrees,acp-mode,skills,headless}.md, gemini-cli/docs/{checkpointing,hooks/index}.md, gemini-cli/docs/tools/*
- qwen-code: qwen-code/README.md, qwen-code/.qwen/skills/
- copilot-cli: copilot-cli/README.md
- amazon-q: amazon-q/docs/{built-in-tools,agent-format,knowledge-management,tangent-mode,introspect-tool,hooks,experiments,todo-lists}.md
- mistral-code: mistral-code/README.md, mistral-code/MISTRAL.md, mistral-code/docs/* (mostly inherited)
- codename-goose: codename-goose/README.md, ARCHITECTURE.md, crates/, documentation/blog/2025-02-21-gooseteam-mcp/index.md
- gptme: gptme/README.md
- open-interpreter: open-interpreter/README.md
- forge: forge/README.md
- gpt-engineer: gpt-engineer/README.md
- swe-agent: swe-agent/README.md, swe-agent/docs/background/aci.md, swe-agent/tools/
- shai: shai/README.md
- nanocoder: nanocoder/README.md, nanocoder/CLAUDE.md, nanocoder/docs/features/*
- claude-squad: claude-squad/README.md
- claude-code: claude-code/README.md, claude-code/plugins/
- vtcode: vtcode/README.md
- warp: warp/README.md, .warp/skills
- get-shit-done: get-shit-done/docs/FEATURES.md, commands/gsd/, AGENTS.md
- spec-kit: spec-kit/README.md
- junie: junie/README.md
- aichat: aichat/README.md
- codai: codai/README.md
- taskweaver: taskweaver/README.md
- octogen: octogen/README.md
- zeroshot: zeroshot/AGENTS.md
- multiagent-coding: multiagent-coding/README.md
- agent-deck: agent-deck/llms.txt
- bridle: bridle/CLAUDE.md, bridle/AGENTS.md, .claude-plugin/marketplace.json (marketplace, not an agent)

HelixCode baseline verified against: helix_code/internal/

{agent,repomap,mcp,plugins,verifier,session,hooks,worker,memory,voice,clar and helix_code/applications/*, plus docs/plans/HXC-031-codex-cline-port.md.